Chapter Notes: Living Creatures: Exploring their Characteristics

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What Makes Living Beings Special?

Living beings show certain key characteristics: they can move, eat, grow, breathe, excrete waste, respond to stimuli, reproduce and finally die. If an entity lacks these features, it is not considered alive.

One sunny morning, Arjun was playing near a pond. He saw a small plant pushing through the soil, a butterfly flying and fish swimming. He wondered why these things move and grow while rocks and water do not. His grandmother explained that living beings have special qualities such as moving, eating, growing and breathing; for example, fish take oxygen from water. Excited to learn more, Arjun set out to discover what makes life special. Are you ready to join him?

What sets Living Things apart from Non-living Things?

Living Things: Living things are beings that show traits like movement, growth, breathing, reproduction, and reacting to their surroundings. Examples include animals, plants, and humans. All living beings move, need food, and grow. They also breathe, reproduce, eliminate waste, react to stimuli, and eventually, die.

Non-living Things: Non-living things are objects that do not have life. They do not show traits like growth, movement, breathing, nutrition, reproduction, or reactions to stimuli. Examples include rocks, water, and toys.

Do plants move from one place to another like animals?

No. Plants do not move their whole body from one place to another like many animals do. However, plants do show certain kinds of movement, for example flowers opening and closing, leaves folding, and climbers wrapping around supports.

Different movements of plants

Plants cannot move from one place to another but they can show various movements. For example, climbers twist around nearby objects to support their growth.

Some plants show movements such as flowers opening in the morning and closing at night, or trapping insects in insectivorous plants.

An insectivorous plant like **Drosera** has saucer-shaped leaves covered with hair-like structures that are sticky at the ends; when an insect lands, these hairs bend inward and trap it.

MULTIPLE CHOICE QUESTION

Try yourself: What is a characteristic feature of plants that sets them apart from animals?

A Plants can move from one place to another like animals.

B Plants cannot absorb water.

C Plants do not require sunlight to survive.

D Plants do not move from one place to another like animals.

View Solution

Growth and Nutrition in Living Beings

All living organisms pass through different stages of life. A tiny seed can grow into a tall tree, and animals develop from young stages to adults. These changes are essential for survival and for producing the next generation.

Growth

Growth means an increase in size, number of cells and complexity. For example, a seed develops into a sapling and then into a tree; animals grow from juvenile stages into adults. Examples include pupae turning into insects and tadpoles becoming frogs.

Nutrition

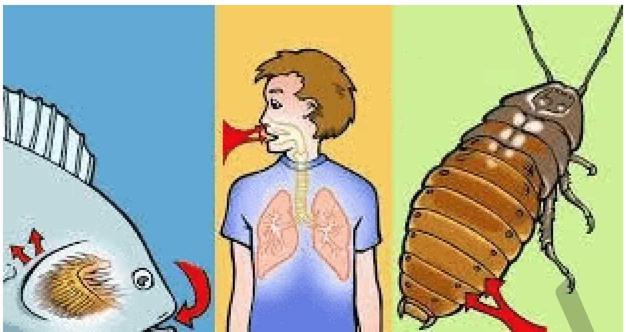
All living beings require **food (nutrition)** to obtain energy for growth, repair and maintaining body processes. Food supplies the materials and energy needed for these activities.

The key characteristics of living beings include movement, eating, growth, breathing, excretion, responding to stimuli, reproduction and death. If any of these features are absent, the organism is not alive.

Let's Revise: How do growth in plants and animals differ or resemble each other?

Ans: Both plants and animals grow by increasing in size and by developing through life stages. For example, a plant grows from a seed to a tree, while animals grow from young to adult. The stages differ by species-tadpoles become frogs, pupae become insects-so the details of growth vary.

Breathing and Respiration in Living Beings



Breathing is the physical process of taking air in (inhaling) and letting it out (exhaling).

Respiration is the biochemical process by which organisms use oxygen to release energy from food.

Plants exchange gases through tiny openings called **stomata**.

All living organisms-plants, animals and microbes-carry out respiration to obtain energy for life.

Let's Revise: What is the difference between breathing and respiration?

Ans: Breathing is the physical act of inhaling and exhaling air, while respiration is the chemical process that uses oxygen to convert food into energy. Respiration occurs within cells of all living beings.

Excretion in Living Beings

Excretion is the process of removing waste products produced by metabolic activities.

Excretion prevents the build-up of harmful substances and helps maintain internal balance. All living beings excrete waste; even plants release some wastes such as excess water and salts.

Excretion in Living Thing

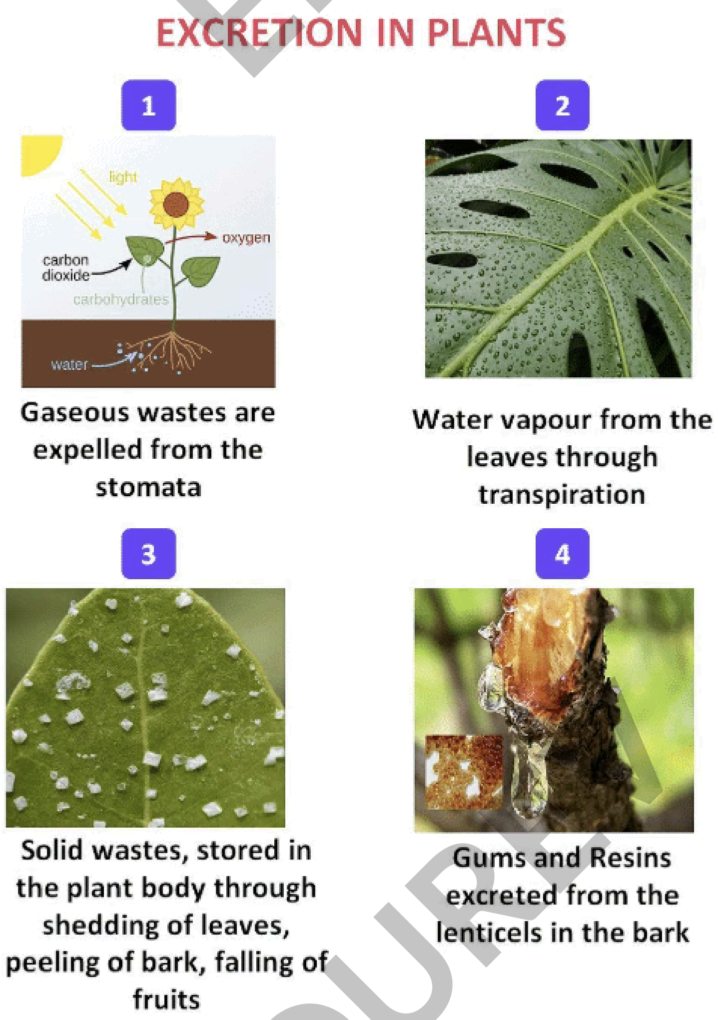
1. Excretion in Animals

Animals use organs such as **kidneys** to filter blood and remove excess water, salts and other wastes, producing **urine**.

Urine is a major excretory product that removes soluble wastes from the body.
Excretion keeps the internal environment stable and safe for cells to function.

2. Excretion in Plants

During **photosynthesis**, plants release oxygen as a by-product; during **respiration** they give out carbon dioxide.
Plants excrete some gases through their stomata and may release excess water and minerals as droplets on their leaves.



Let's Revise

Q: What are some examples of waste products excreted by plants and animals?

Solution:

Ans: In animals, common waste products include urine, which contains water, salts, and other substances filtered from the blood. In plants, waste products include oxygen (from photosynthesis), carbon dioxide (from respiration), and excess water and minerals (released as droplets on leaves).

Q: How do animals and plants differ in their method of excretion?

Solution:

Ans: Animals use organs like kidneys to filter waste from the blood and produce urine, which removes water, salts, and other waste materials. Plants, on the other hand, release gases like oxygen and carbon dioxide through stomata and may excrete excess water and minerals as droplets from their leaves.

Response to Stimuli

A **stimulus** is any change in the environment that causes a living being to react. For example, touching a hot object makes you quickly pull your hand away. All living things respond to stimuli, but the speed and type of response vary.

Response to Stimuli in Living Beings

Plant Responses

- Plants respond to stimuli like touch, light and changes in temperature. For example, the touch-me-not (mimosa) plant folds its leaves when touched.
- Certain plants change their leaf positions at night or in response to bright light; for example, some species fold leaves after sunset.

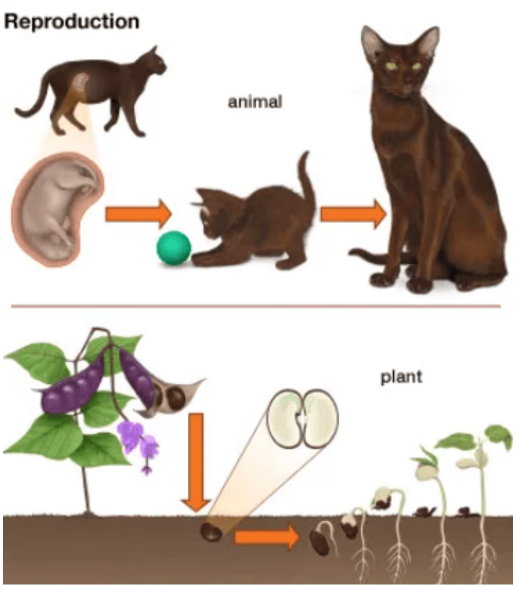
Let's Revise: How do plants respond to stimuli? Give examples.

Ans: Plants respond to touch, light and environmental changes. For example, the touch-me-not closes its leaves when touched; other plants fold leaves after sunset. These movements occur even though plants lack a nervous system; they use chemical and physical changes to respond.

Reproduction in Living Beings

Reproduction is the process by which living organisms produce new individuals of the same kind.

Reproduction is essential for the survival of species and continuation of life on Earth.

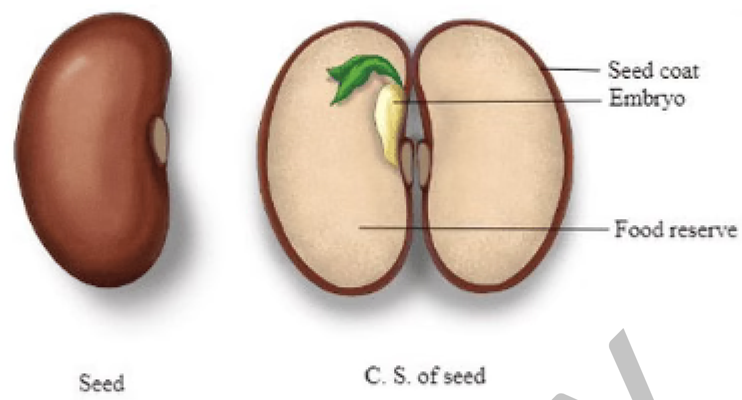


Reproduction in Living Beings

A living being is considered dead when it no longer shows signs of life such as movement, growth or the need for food even when provided with suitable conditions like air, water and food.

Essential Conditions for Germination of a Seed

Why is water important for seed germination?
Water activates enzymes inside the seed that begin the growth process. It softens the seed coat so the embryo can emerge and supports the biochemical reactions needed for development.



Germination is the process in which a seed grows into a new plant when conditions are suitable.

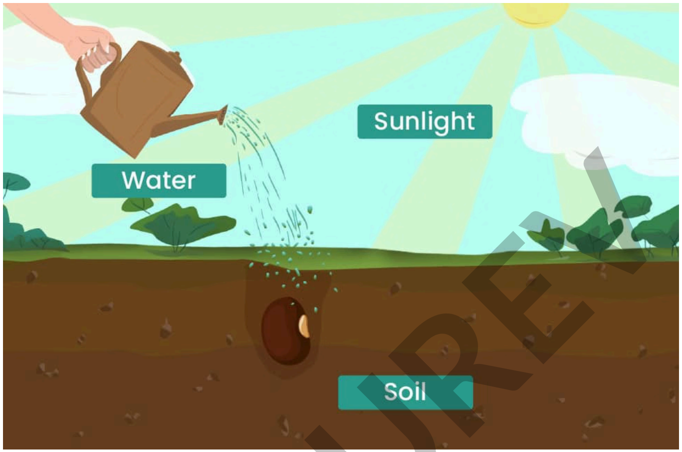
Essential conditions: Water to activate growth processes and soften the seed coat; air (oxygen) for respiration; and the right temperature. Most seeds do not require light to begin germination, but seedlings need sunlight after germination for photosynthesis.

Example: Bean seeds absorb water, the seed coat splits and the radicle (young root) and plumule (young shoot) emerge to form a seedling.

Water: Seeds need water to start the chemical reactions that lead to growth.

Air and soil: Oxygen from air in the soil is required for respiration; soil structure with air spaces also helps roots grow.

Light and dark conditions: Many seeds (for example, bean seeds) do not need light to germinate, but some seeds require light or darkness-see below.



Do you know?

Some seeds of flowering plants like **Coleus** and **Petunia** require light to begin germinating; if buried too deeply, they will not sprout. Other seeds, such as **Calendula** and **Zinnia**, require darkness and should be covered by soil to germinate well.

Growth and Movement in Plants

Upright position: The **root** grows downward and the **shoot** grows upward.

Inverted position: If a plant is turned upside down, the root bends to grow downward and the shoot bends to grow upward.

Directional sunlight: If light comes from one side, the shoot grows towards the light (phototropism) while the root continues to grow downward.

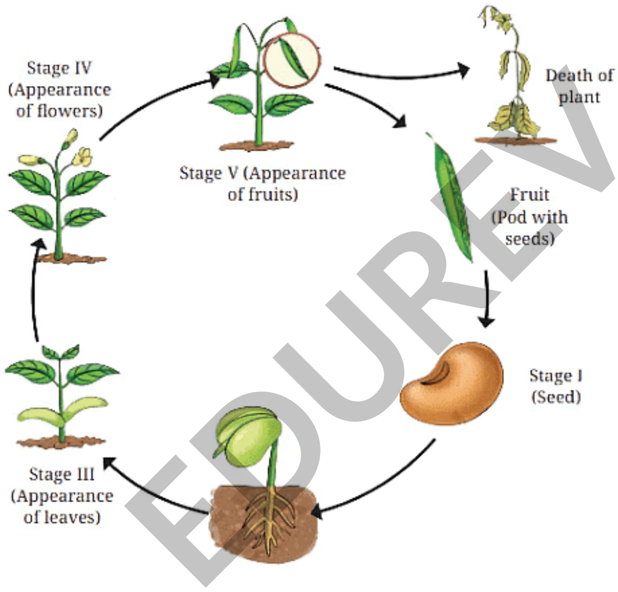
Life Cycle of Plants

Seed germination: A seed sprouts and becomes a young plant when conditions are suitable.

Growth and maturation: The young plant grows, produces flowers and then fruit. Fruits often contain seeds.

Seed production: Seeds from fruits give rise to the next generation. The sequence from seed to plant to seed is the plant's **life cycle**.

Plant death: When a plant stops growing and life functions cease, it is dead.

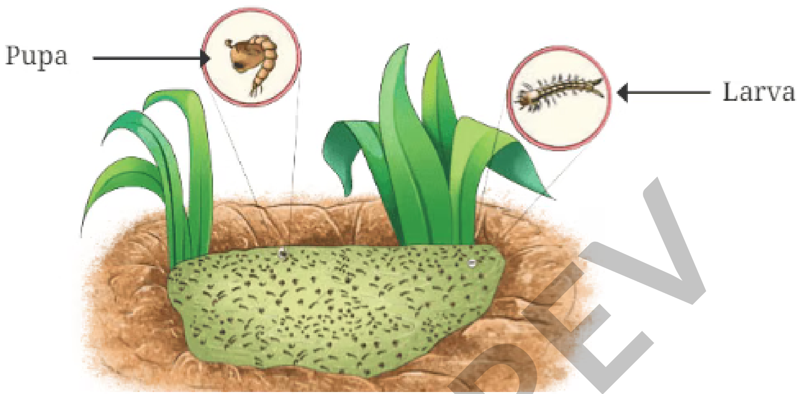


Life Cycle of Animals

Do all animals have the same life cycle?

No. Different animals have different life cycles. Some undergo metamorphosis (for example butterflies), while others develop more directly (for example humans).

Life Cycle of a Mosquito



Egg: Laid on or near water.

Larva: Lives in water and feeds on organic matter.

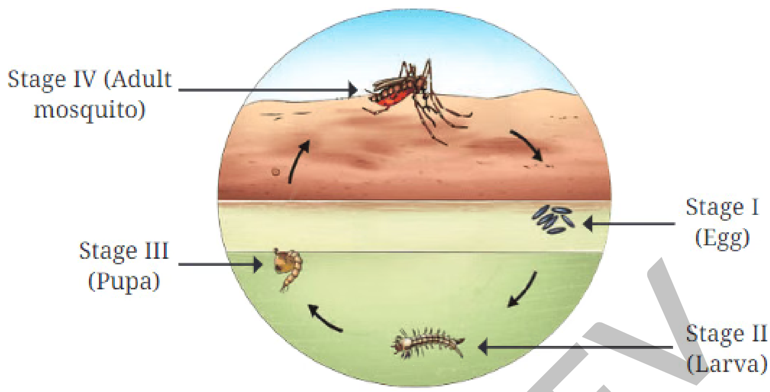
Pupa: A resting stage; does not eat.

Adult: Emerges from pupa, can fly and reproduce.

The adult rests briefly on the water surface before flying away. Adult mosquitoes typically live about 10-15 days.

The female lays eggs on or near water, repeating the cycle.

Each stage (egg → larva → pupa → adult) differs clearly in appearance and body structure.



Life Cycle of a Mosquito

Do you know?

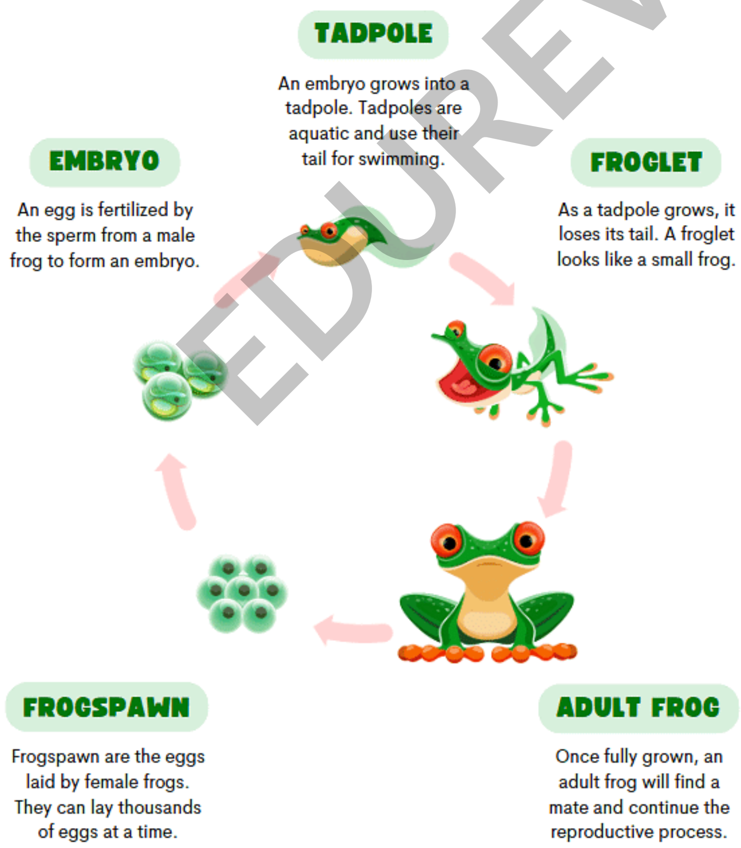
The silk moth goes through four stages: egg, larva, pupa and adult.

Larvae (caterpillars) grow and produce thread-like silk that they use to form cocoons, leading to the pupal stage.

The silk threads from these larvae are the raw material for silk fabric.

In India, the Khadi and Village Industries Commission (**KVIC**) supports silk production and related small industries.

Life Cycle of a Frog



Life Cycle of a Frog

Egg (spawn): Laid in water, usually in clusters.

Embryo: The developing stage inside the egg.

Tadpole: Aquatic young with tails and gills; later develop hind legs.

Froglet: Transitional stage-has legs and a shrinking tail; begins to live partly on land.

Adult frog: Fully developed and able to live on land and in water; can reproduce.

Duration of Each Stage (typical example)

1. Stage I - Spawn (Day 1)
2. Stage I - Embryo (Day 7-10)
3. Stage II - Tadpole with legs (8-10 weeks)
4. Stage III - Froglet (about 12 weeks)
5. Stage IV - Adult frog (about 14 weeks)

Key Points

- We encounter both **living beings** and **non-living things**; understanding their differences is fundamental in biology.
- Living beings show movement; animals have obvious movement, whereas plants show movements like **phototropism** and other growth movements.
- **Growth** means an increase in size and number of cells and depends on nutrition.
- Living beings need **food** for energy, growth and repair.
- **Gas exchange** (oxygen and carbon dioxide) is essential for cellular energy production.
- Removing metabolic wastes by **excretion** maintains internal balance.
- Living beings respond to environmental changes for protection and adaptation.
- **Reproduction** ensures the continuation of species.
- If an organism cannot show life processes even in suitable conditions, it is considered **dead**.
- Each organism passes through a series of stages in its life; seed germination depends on water, air and suitable light conditions for the species.

MULTIPLE CHOICE QUESTION

Try yourself: Which of the following animals does not go through a larval stage in its life cycle?

A Mosquito

B Silk Moth

C Butterfly

D Earthworm

View Solution